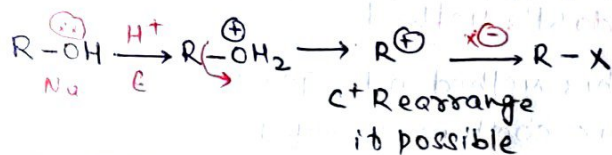


HALOALKANES AND HALOARENES

MOP of Haloalkane

→ ① from Alcohols

Rxn with HX: $(H-X \rightarrow H^+ + X^-)$



$R-OH \propto$ stability of 1st carbocation

→ Luca's Test for Alcohols

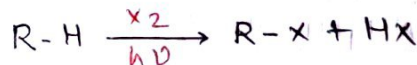
Lucas reagent :- conc. HCl + Anhy. ZnCl₂

Normal (without Re₂O)

- 1° ROH gives no turbidity / Long time
- 2° ROH → turbidity (in minutes)
- 3° ROH → turbidity (in seconds)

* Re₂O stabilized 1° c⁺, 2° c⁺, 3° c⁺, turbidity in seconds.

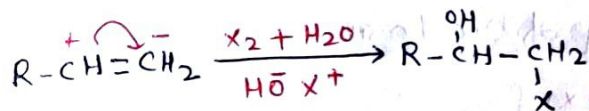
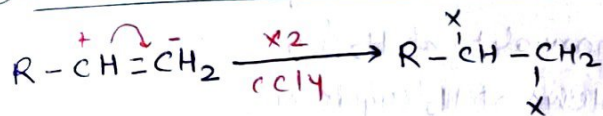
② From Halogenation of Alkanes



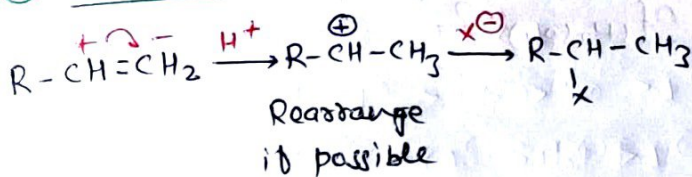
- Total no. of mono halogenated products.
- Major $\begin{cases} Cl \rightarrow n^{\circ}, n^{\circ}, n^{\circ}, n^{\circ} \\ Br \rightarrow 3^{\circ} > 2^{\circ} > 1^{\circ} \end{cases}$

$$r_p : r_s : r_t = 1 : 3.8 : 5$$

③ from Halogenation of Alkenes

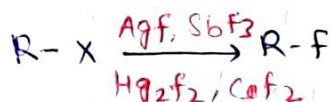


④ from hydrohalogenation of Alkenes



⑤ From Halogen exchange

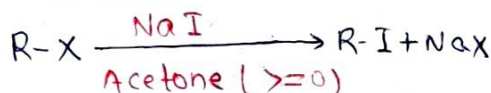
① Swartz Reaction :- (Inorganic blonides)



(X = Cl, Br, I)

X hatao
F lagao

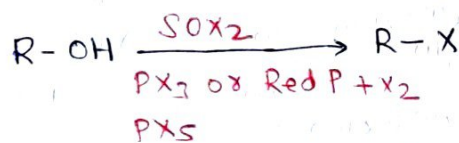
② Finkelstein Reaction



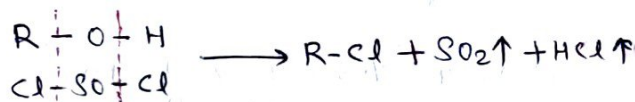
(X = Cl, Br) or DMSO (dimethyl sulfoxide)
> S = O

X hatao
I lagao

③ From Alcohol without rearrangement

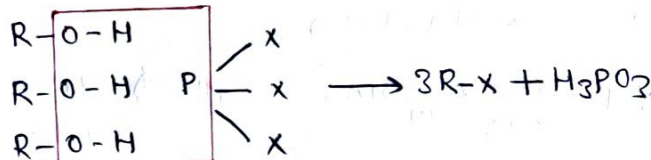


① Darzen's rxn :- (Best M.O.P of R-Cl)

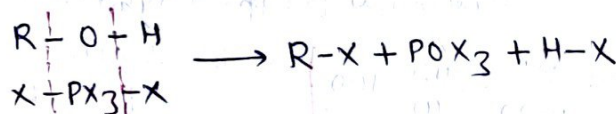


(SOCl₂ thionyl chloride)

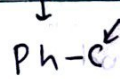
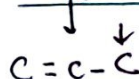
② 'P X₃' or 'Red P + X₂' (X = Cl, Br)



③ P X₅ (X = Cl, Br)



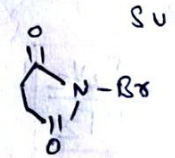
⑦ Allylic or Benzylic substitution

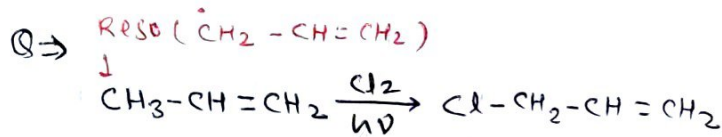


chlorination

- Cl₂/hν (Cl[•])
- Cl₂/500°C (Cl[•])
- t-buo-Cl (Cl[•])
(→ o-cl)
- Cl₂ (low concⁿ)/hν (Cl[•])

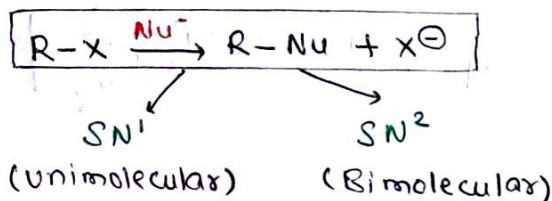
Bromination

- Br₂/hν (Br[•])
- NBS (N-bromo Succinamide)
 (Br[•])
- Br₂ (low concⁿ)/hν (Br[•])



Properties of Haloalkanes

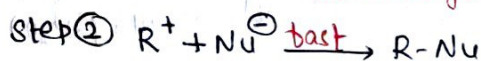
→ NSR (Nucleophilic substitution Rxn)



(a) SN^1 Rxn:-

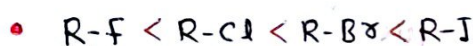


Rearrangement is pos.

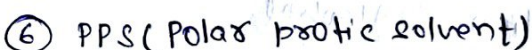
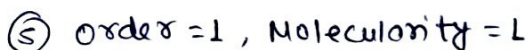
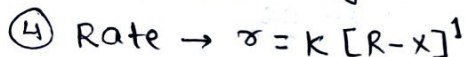


→ OP Points of SN^1

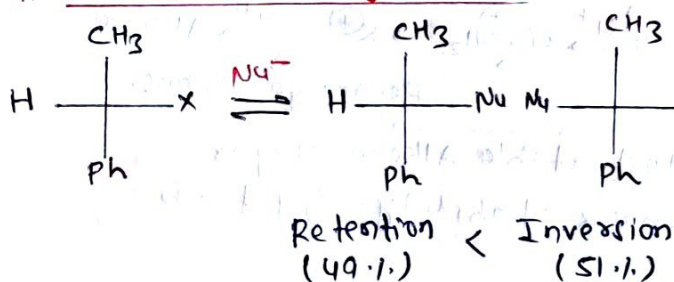
- ① Carbocation intermediate formation
- ② 1st step is the R.D.S (slow step)
- ③ ④ R.O.R w.r.t Halogen



⑤ R.O.R w.r.t Alkyl

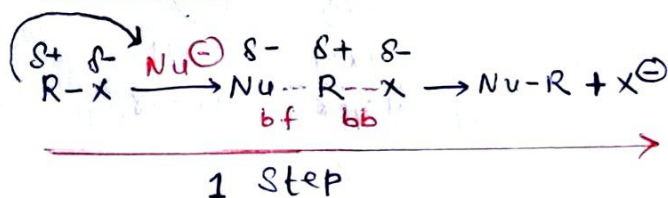


Stereochemistry of SN^1



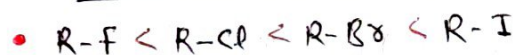
Enantiomer pair 50% Reten + 50% Inv
100% Racemization

(b) SN^2 Rxn:-

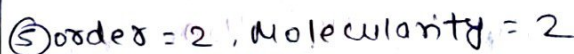
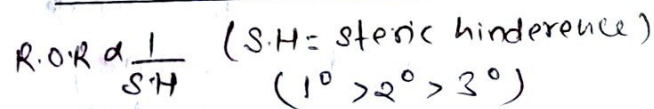


→ OP Points of SN^2

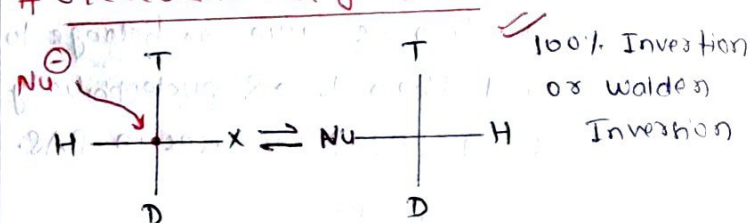
- ① Transition state formation
- ② 1st step is the R.D.S (slow step)
- ③ ④ R.O.R w.r.t Halogen



⑤ R.O.R w.r.t Alkyl

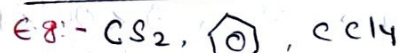


Stereochemistry of SN^2



Effect of solvent:-

→ Non-polar solvent :- $\text{D.N} = 0$

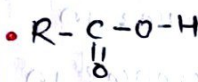
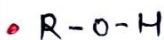
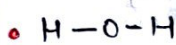


→ Polar solvent :- $\text{D.N} \neq 0$

Polar protic solvent (PPS)

Polar Aprotic solvent (PAS)

'O' se 'H' juda hoga 'O' se 'H' nahi juda hoga

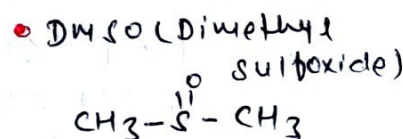
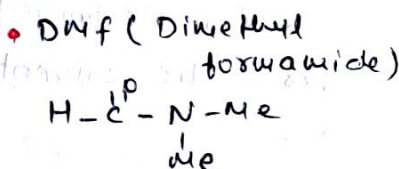
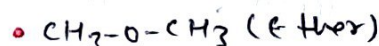
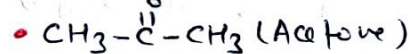


→ PPS favours SN^1 rxn.

↓
solvation or hydration

→ $\text{Nu}^- \rightarrow$ weak
↓
charge density ↓

generally $\leftarrow \text{Nu}^- \rightarrow$ strong (-ve charge)
(-ve > neutral)

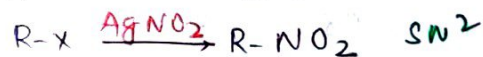
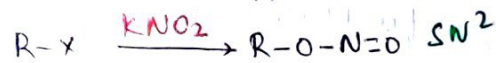
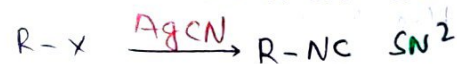
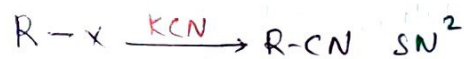


→ PAS fav SN^2 rxn

→ ① cation solvation

→ ② Anion repulsion (Nu)

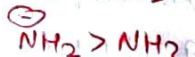
Most important rxn



Nucleophilicity:- Tendency to donate e^-

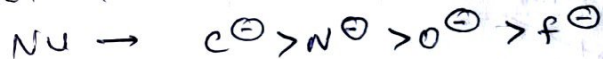
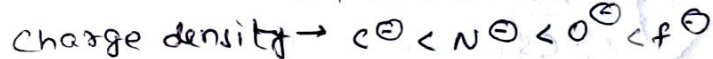
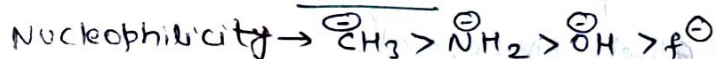
- ① Negatively charged nu is more nucleophilic than corresponding neutral atom.

charge density
-vely > neutral

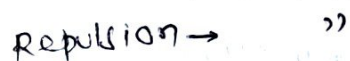
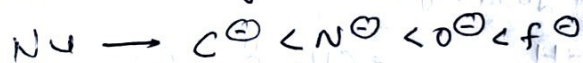
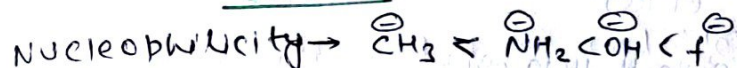


- ② If donor atom is same or belongs to same period then $L \rightarrow R$ nucleophilicity decreases in PPS & increases in PAS.

In PPS

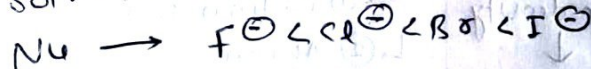
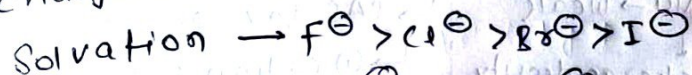
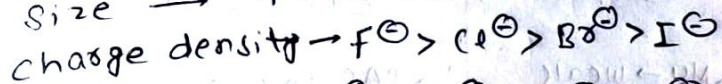
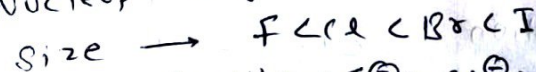
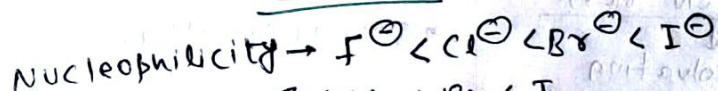


In PAS

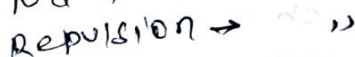
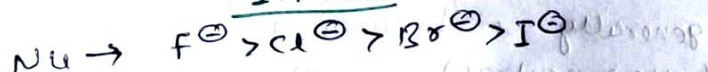


- ③ $T \rightarrow B$ nucleophilicity increases in polar protic solvent & decreases in PAS.

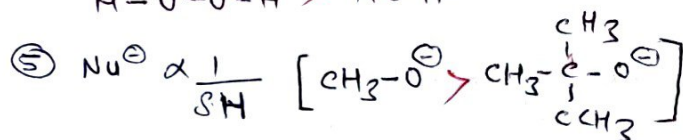
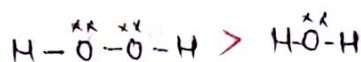
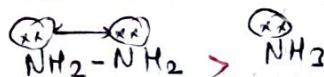
In PPS



In PAS

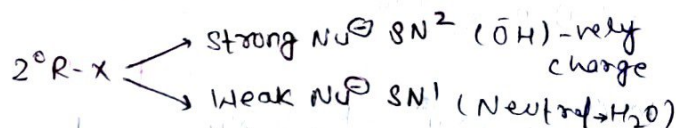
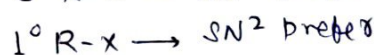


- ④ If L.P containing atom present adjacent to nu^- atom then it will be more nu^- .



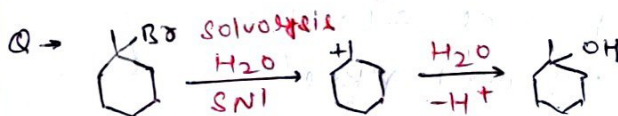
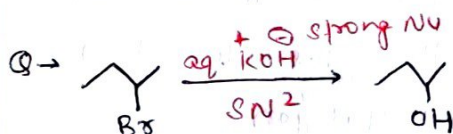
OP Point:-

- ① $3^\circ R-X \rightarrow S_N1$ prefer



- ② Lewis Acid $\rightarrow S_N1$ prefer

- ③ Solvolysis $\rightarrow S_N1$ prefer

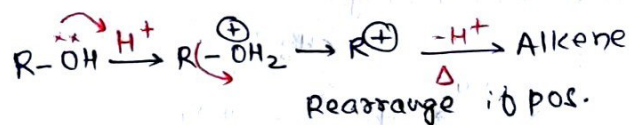


Elimination Reactions

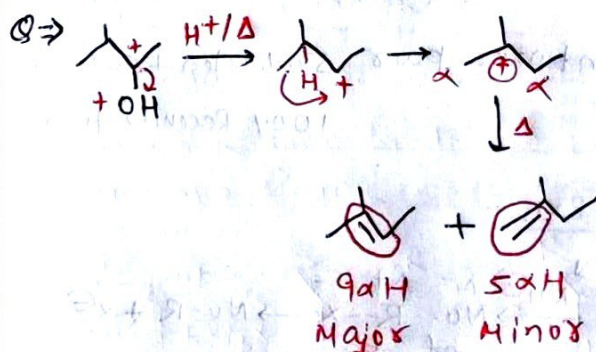
① E₁ Rxn

① Dehydration of Alcohol

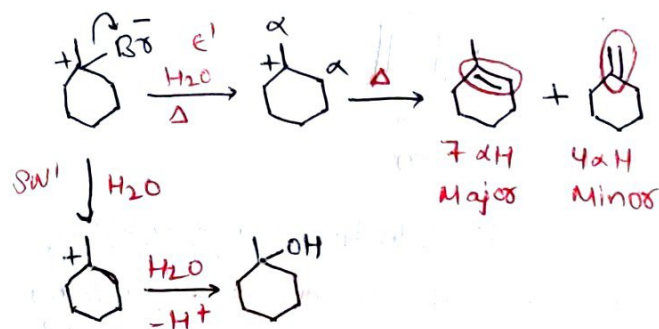
Reagent:- H^+/Δ or Conc. H_2SO_4/Δ
or Conc. H_3PO_4/Δ



Most stable Alkene Major
ROR $\propto$ stability of 1st e^-

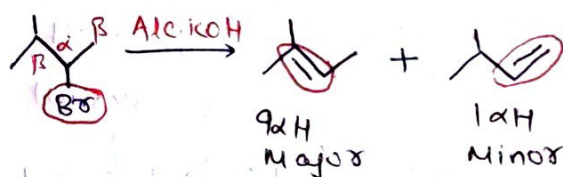


⑥ Heating of R-X in PPS



② E² Rxn :-

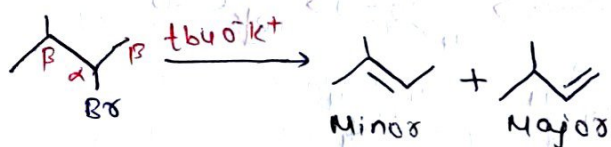
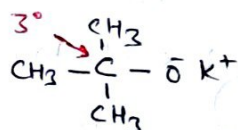
Reagent:- Alk KOH or RO⁻Na⁺



1. More stable alkene Major
2. NO rearrangement
3. ROR α stability of alkene
3° > 2° > 1°

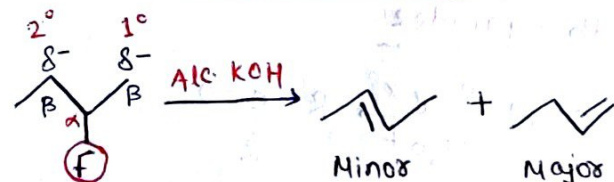
Special case

When bulky base is present



Less hindered β will give major product.

③ E^{2a} (Anion like character)



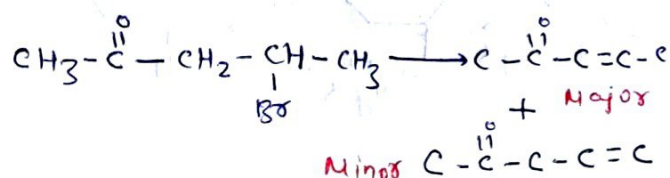
C-F
Bond strong

1° C⁻ > 2° C⁻
1° C^{δ-} > 2° C^{δ-}

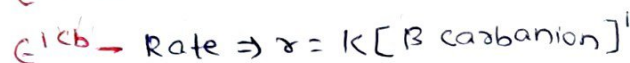
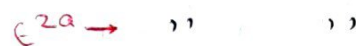
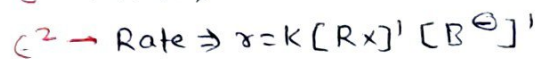
More stable β C^{δ-} will decide Major pdt.

④ E^{1cb} (conjugate base formation)

When β C⁻ is stabilized by reso.

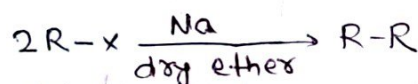


More stable β C⁻ will decide Major

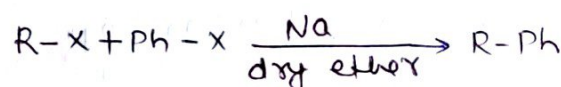


Reaction with Metals

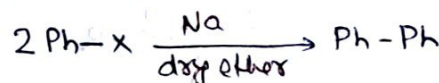
① Wurtz Reaction :-



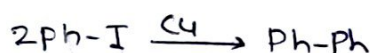
② Wurtz-Fittig Reaction :-



③ Fittig Reaction :-

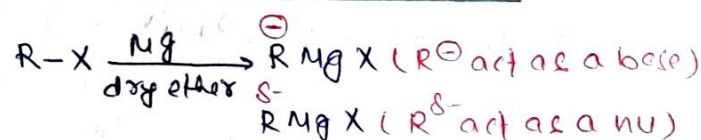


④ Ullmann Reaction :-



GRIGNARD REAGENT

MOP of Grignard Reagent



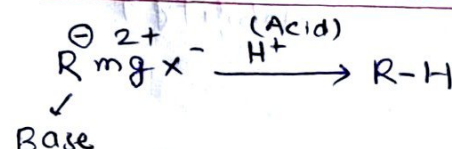
OP Points:-

- ① ROR → R-I > R-Br > R-Cl > R-F
- ② In moist condition alkane will be formed.

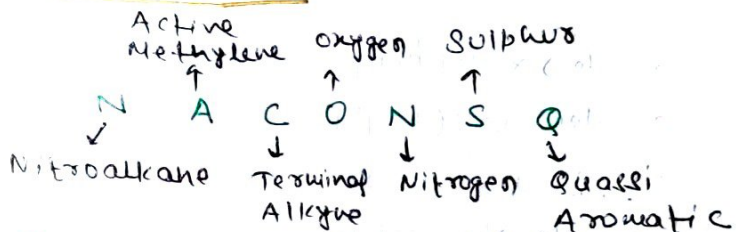
$$\text{R}^-\text{Mg}^+\text{X}^+ \xrightarrow{\text{H}^+/\text{OH}^-} \text{R-H} + \text{Mg(OH)X}$$
- ③ G.R is an organometallic comp → comp in which metal is directly bonded with carbon.
- ④ Dry ether stabilizes G.R & decreases the reactivity of G.R.

Chemical Reactions :-

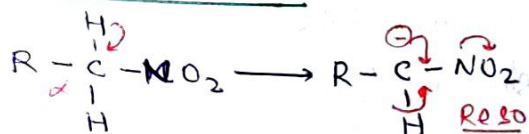
Acid base rxn or zewittinoff's rxn



* OP TRICKS

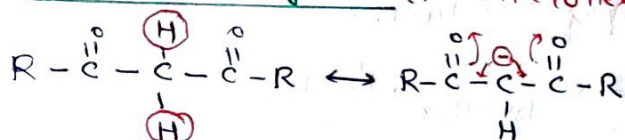


① Nitroalkanes

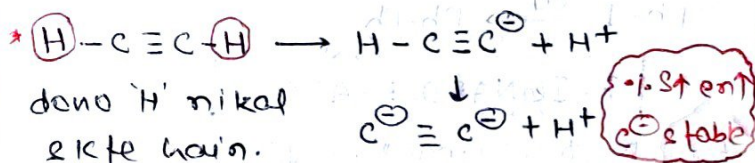


Na waba hogi na waba ki baat hogi, ek he atom par ek se jyada 'H' honge toh rxn ek he bar hogi.

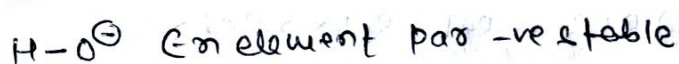
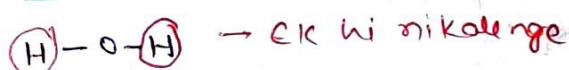
② Active Methylene (β-diketones)



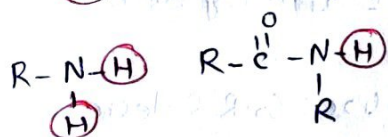
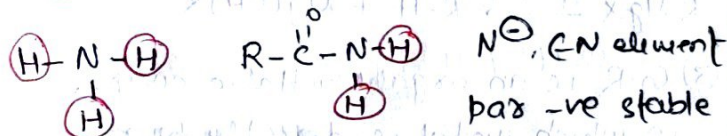
③ Terminal alkynes



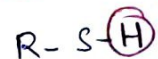
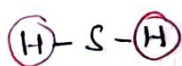
④ Oxygen



⑤ Nitrogen

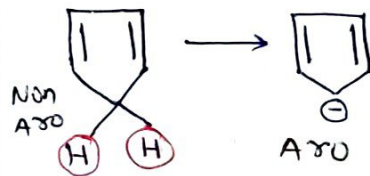


⑥ Sulphur



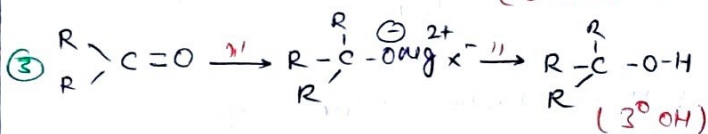
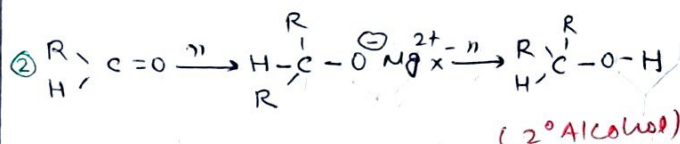
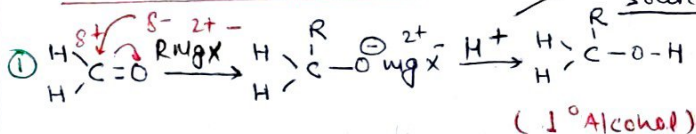
$\text{S}^\ominus$ size ↑
 stability ↑

⑦ Quasi-aromatic

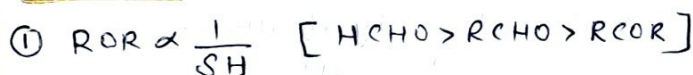


Nucleophilic Addition Rxn (NAR):-

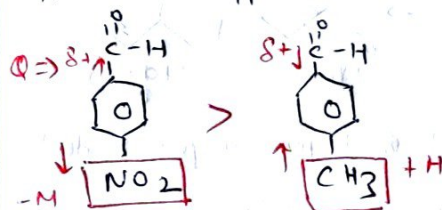
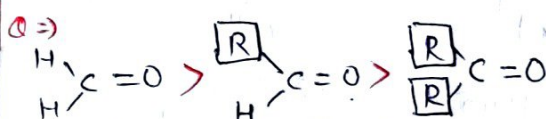
① With carbonyl compounds:- $\xrightarrow{\text{H}_2\text{O}/\text{NH}_4\text{Cl source}}$



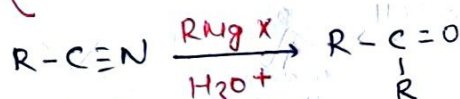
OP Points:-



② Electronic factor:-



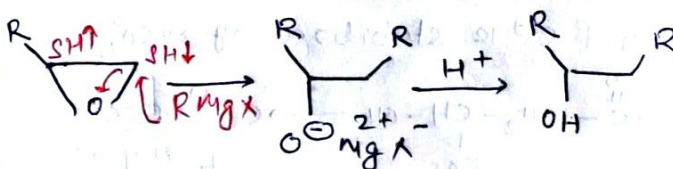
② With cyanides



G.R ke R cyanides 'C' se jod kar ketone bana denge

③ With cyclic ether:- (only 3 & 4 memb)

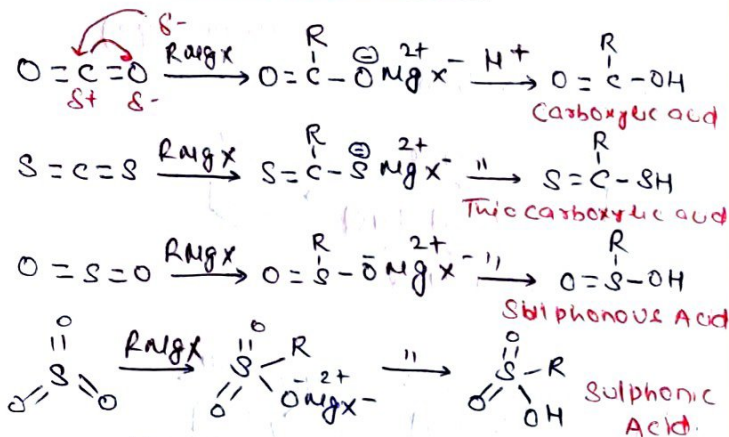
G.R ka R less hind 'C' se jod kar OH bana denge



* OP Points :-

- Acyclic ether does not react with Gr.
- 3 & 4 membered cyclic ether can react with Gr.
- 5 membered & beyond do not react with Gr.

④ With oxides & Sulphides

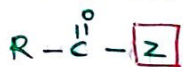


Gr. R ka 'R' central atom se jod kar Acid bana denge.

⑤ Nucleophilic Addition elimination rxn

(Rxn with Acid derivatives)

(S_NAc)



- ① $\text{R}-\text{C}(=\text{O})-\text{Cl}$ Acid chloride, $\text{Z}=\text{Cl}$
- ② $\text{R}-\text{C}(=\text{O})-\text{OCOR}$ Anhydride, $\text{Z}=\text{OCOR}$
- ③ $\text{R}-\text{C}(=\text{O})-\text{OR}$ Ester, $\text{Z}=\text{OR}$
- ④ $\text{R}-\text{C}(=\text{O})-\text{NR}_2$ 3° Amide, $\text{Z}=\text{NR}_2$

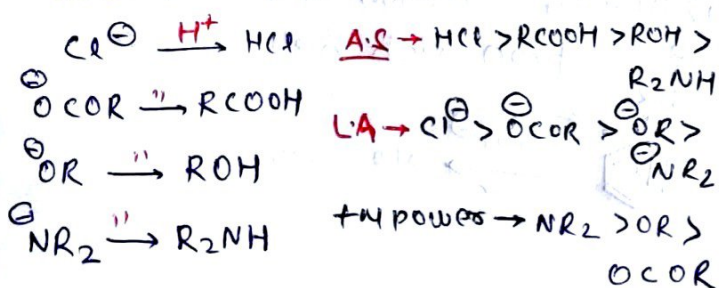
S_NAc Mechanism:-



Z hatao, Gr. R ka R lagao

R.O.R (rate of rxn) or reactivity in S_NAc

- ① Leaving Ability:- Stronger conjugate acids (or weak bases) are better L.G.



② Electronic factors:-

ROR & electrophilicity & EWG $\frac{\alpha}{\text{edge}}$



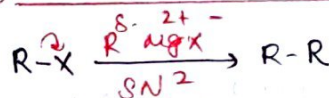
* OP points:-

- ① Acid base rxn are fastest

- ② NAR & S_NAc

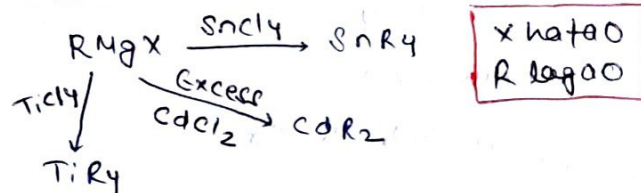
ROR $\rightarrow$ Acid chlorides > Anhydrides > Aldehydes > Ketones > esters > Amides.

- ③ Nucleophilic substitution rxn

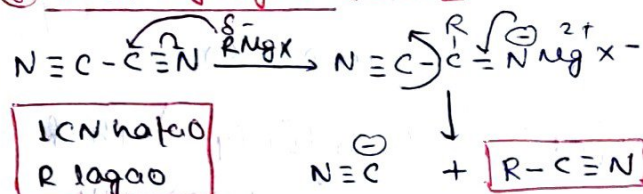


X hatao, Gr. R ka R lagao

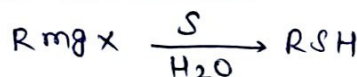
- ④ With Pld block halides



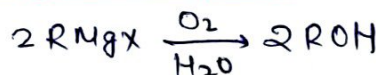
- ⑤ with cyanogen (CN)₂



- ⑥ with sulphur:-



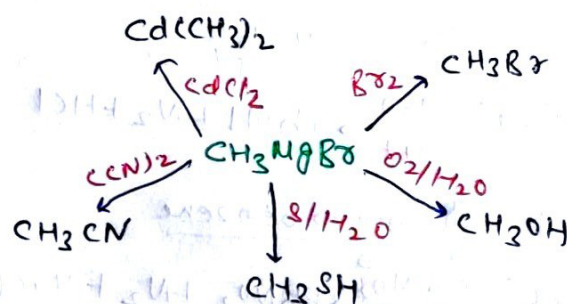
- ⑦ with oxygen



- ⑧ with Halogens

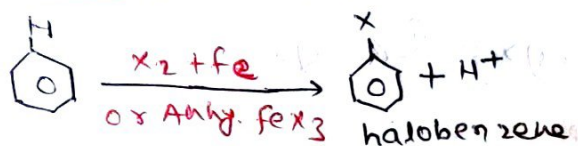


Q=



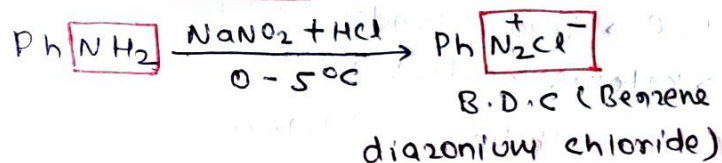
MOP of Haloarenes

① ESR ($X = Cl, Br$)

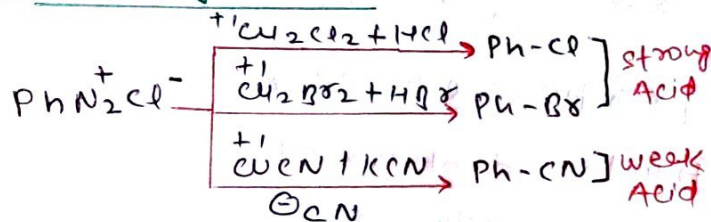


H katao, X logao

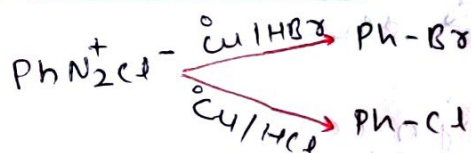
② from Aniline



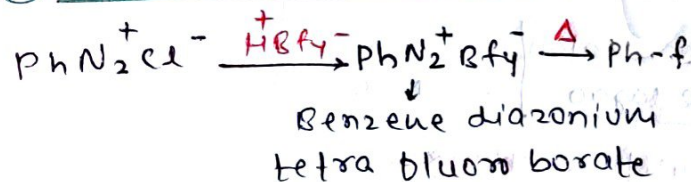
③ Sandmeyer rxn



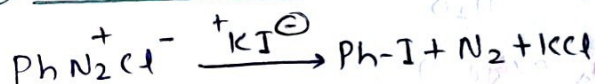
④ Gattermann Rxn



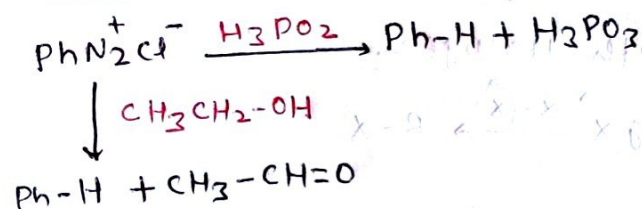
⑤ Balz-Schiemann Rxn



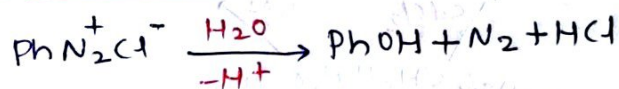
⑥ Sizzzi's Rxn



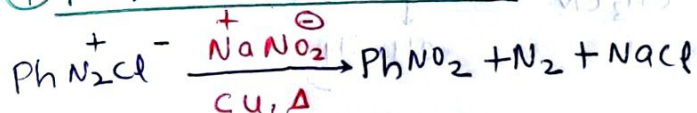
⑦ Reduction Rxn



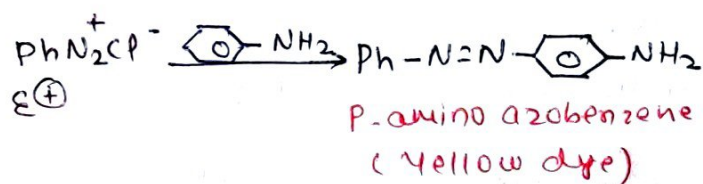
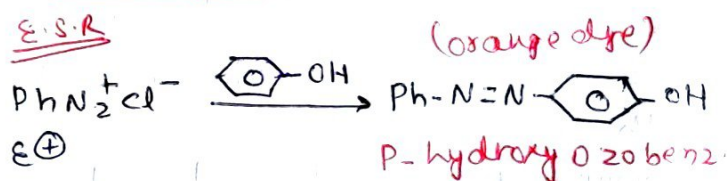
⑧ Rxn with warm H₂O



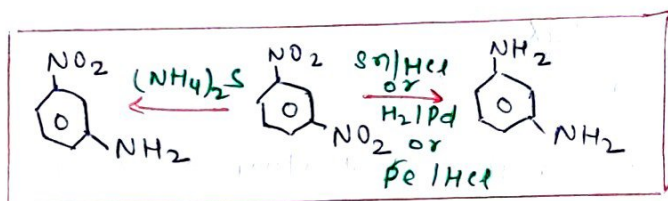
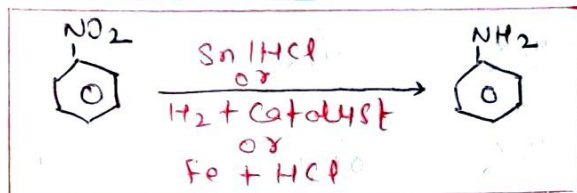
⑨ Prep. of nitrobenzene



⑧ Coupling Rxn:-



* Other Reactions

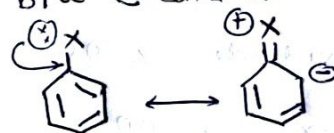


Properties of Haloarenes

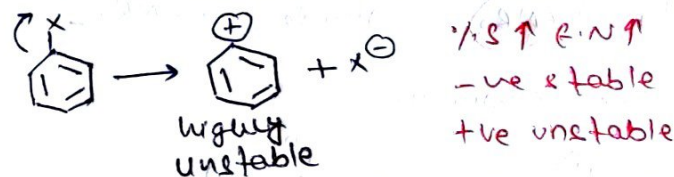
Generally Aromatic N.S.R do not occurs.

Reasons

① Due to partial double bond character b/w C and X.



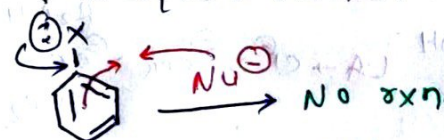
② Due to instability of phenyl cation



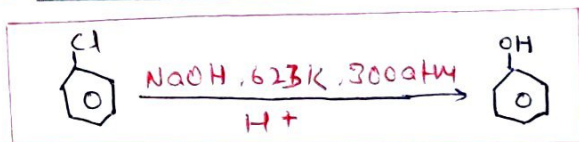
③ Due to difference in hybridization of carbon of C-X bond.



④ Due to Repulsion b/w nucleophile & electron rich haloarenes.



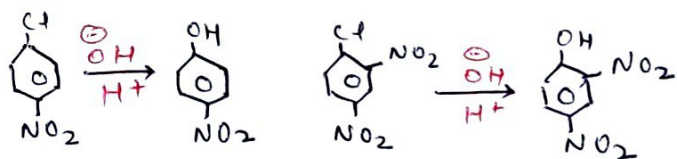
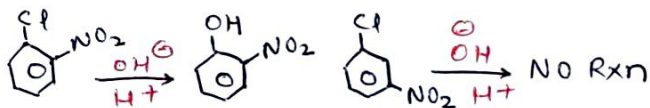
① Dow's process (Ar N.S.R)



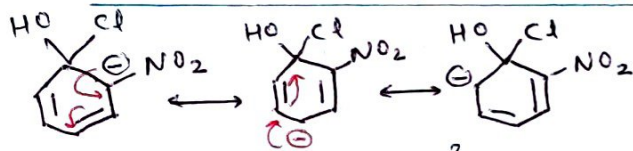
② Ar N.S.R (Ar SN²)

① In the presence of EWG (electron withdrawing group) like Nitro N.S.R.

② O, P nitrohalobenzene can give N.S.R



Meisenheimer complex - o-nitro



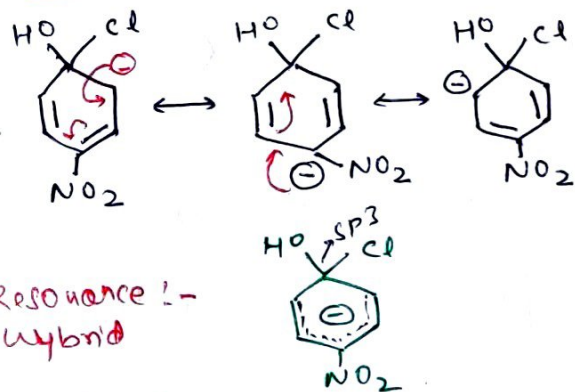
Resonance hybrid

or Meisenheimer complex

E.S.R → sigma complex or wheeland inter

Ar. N.S.R → Meisen [⊖] complex

Meisenheimer complex - p-nitro



Resonance :- hybrid

* OP Points :-

① Also known as Ar SN².

② Carbanion intermediate formation

③ 2 step rxn.

④ 1st step is R.D.S (formation of [⊖])

⑤ Rate → $r = k [O, P \text{ nitro halobenzene}] [Nu^{\ominus}]$

⑥ Bimolecular rxn, order = 2
Molecularity = 2.

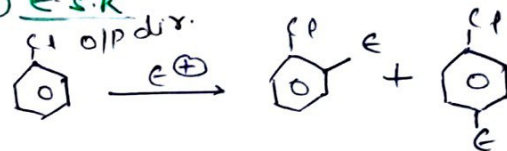
⑦ R.O.R α no. of nitro groups at O & P

If NO₂ groups same → O > P

⑧ R.O.R w.r.t Halogen

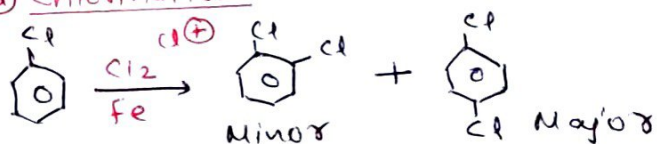
-I power → F > Cl > Br > I

③ E.S.R

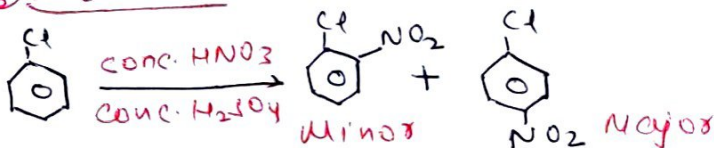


[Always para major (S.H factor)]

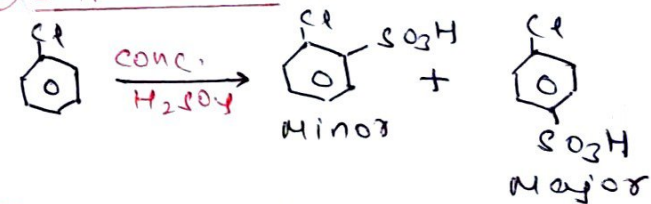
① chlorination



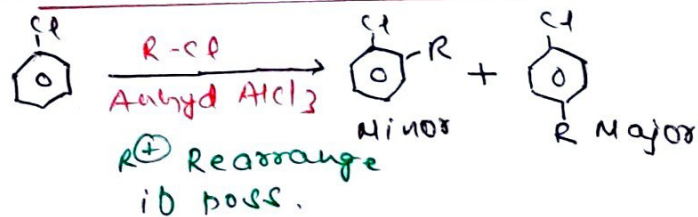
② Nitration



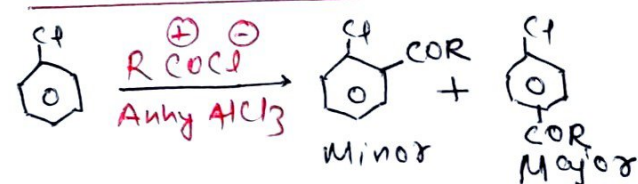
③ Sulphonation



④ Friedel craft Alkylation

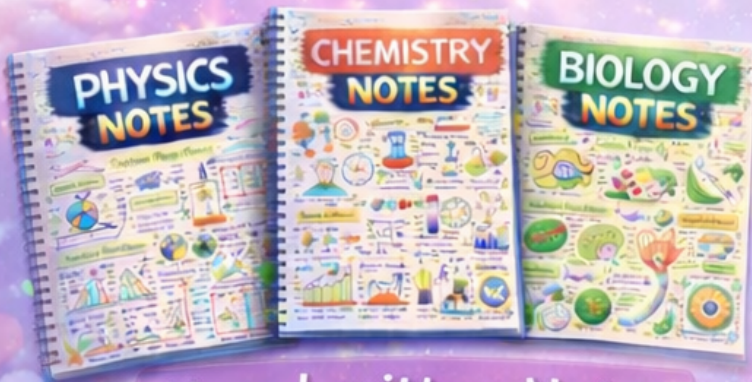


⑤ Friedel craft Acylation



By @THUNDERSTUDY

Other By @Thunderstudy



Handwritten Notes

By Pragma



Boards Dominator Series

By @Thunderstudy

Best Notes Best Book
For PCMB

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